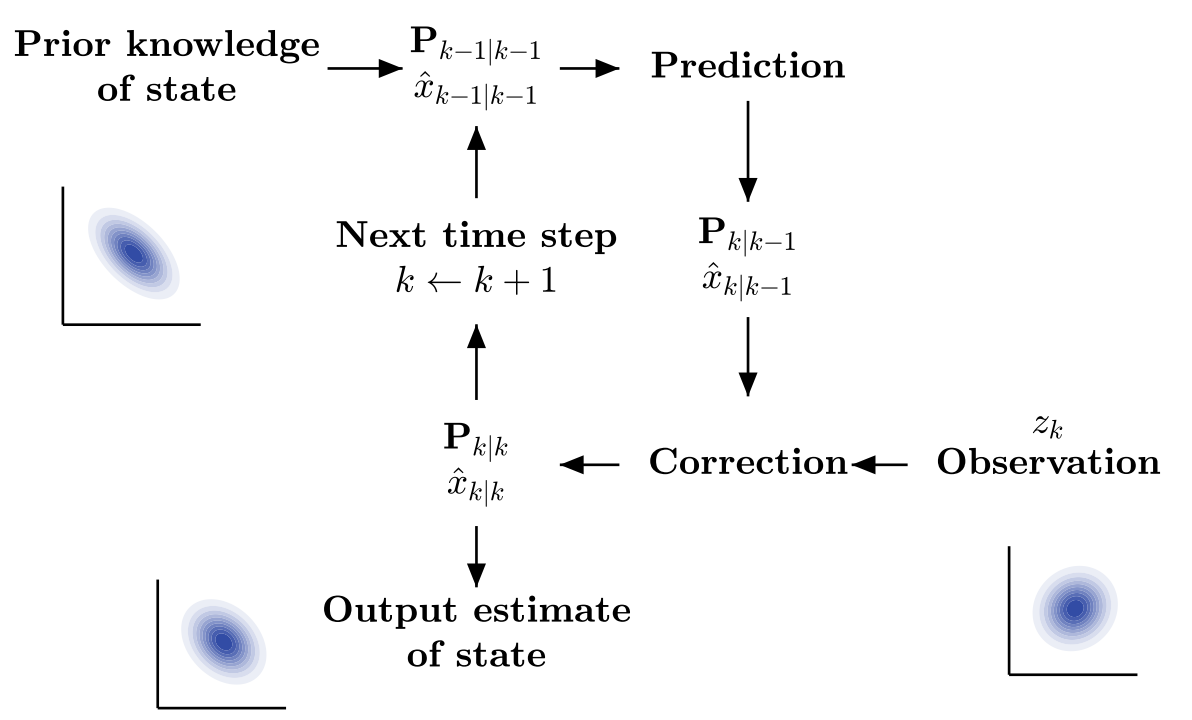

Prior knowledge of state
$\mathbf{P}_{k-1|k-1}$
$\hat{x}_{k-1|k-1}$
Prediction
$\mathbf{P}_{k|k-1}$
$\hat{x}_{k|k-1}$
Next time step
$k \leftarrow k+1$
$\mathbf{P}_{k|k}$
$\hat{x}_{k|k}$
Correction
$z_k$
Observation
Output estimate of state